## To find the problem statement in Assignment-Regression Algorithm

## I. Client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

**1. MULTIPLE LINEAR REGRESSION** ($R^2$ score value**)** = 0.7680

**2. SUPPORT VECTOR MACHINE:**

| S.NO | Hyperparameter | LINEAR (r value) | RBF (NON LINEAR) (r value) | POLY (r value) | SIGMOID (r value) |
|---|---|---|---|---|---|
| 1 | C = 10 | 0.4920 | -0.0393 | 0.0438 | 0.0420 |
| 2 | C = 100 | 0.6447 | 0.3391 | 0.6514 | 0.5409 |
| 3 | C = 500 | 0.7619 | 0.6953 | 0.8476 | 0.4948 |
| 4 | C = 1000 | 0.7343 | 0.8235 | 0.8675 | 0.2221 |
| 5 | C = 2000 | 0.7328 | 0.8662 | 0.8671 | -0.7144 |
| 6 | C = 3000 | 0.7327 | 0.8778 | 0.8667 | -3.1313 |
| 7 | C = 5000 | 0.7313 | 0.8854 | 0.8668 | -8.4400 |

The **SVM regression** use $R^2$ value (Non-linear(rbf) and hyper parameter (C5000)) = 0.8854

**3. DECISION TREE:**

| S.NO | CRITERION | MAX_FEATURES | SPLITTER | R VALUE |
|---|---|---|---|---|
| 1 | squared_error | sqrt | best | 0.7276 |
| 2 | *squared_error* | log2 | best | 0.7276 |
| 3 | squared_error | sqrt | random | 0.6714 |
| 4 | squared_error | log2 | random | 0.6714 |
| 5 | *absolute_error* | sqrt | best | 0.7668 |
| 6 | *absolute_error* | log2 | best | 0.7668 |
| 7 | *absolute_error* | sqrt | random | 0.7920 |
| 8 | *absolute_error* | log2 | random | 0.7920 |
| 9 | *poisson* | sqrt | best | 0.7078 |
| 10 | *poisson* | log2 | best | 0.7078 |
| 11 | *poisson* | sqrt | random | 0.6551 |
| 12 | *poisson* | log2 | random | 0.6551 |

**BEST MODEL WHILE HYPERTUNING IN DECISION TREE FOR PROBLEM**

| S.NO | CRITERION | MAX_DEPTH | MIN_LEAF | MIN_SPLIT | SPLITTER | R VALUE |
|---|---|---|---|---|---|---|
| 1 | absolute_error | 7 | 5 | 7 | best | 0.8887 |

The **Decision Tree regression** use $R^2$ value = 0.8887

**4.RANDOM FOREST:**

| S.NO | N_ESTIMATORS | CRITERION | MAX_FEATURES | R VALUE |
|---|---|---|---|---|
| 1 | 10 | squared_error | sqrt | 0.8489 |
| 2 | 10 | squared_error | log2 | 0.8441 |
| 3 | 50 | squared_error | sqrt | 0.8674 |
| 4 | 50 | squared_error | log2 | 0.8682 |
| 5 | 100 | squared_error | sqrt | 0.8708 |
| 6 | 100 | squared_error | log2 | 0.8716 |
| 7 | 10 | absolute_error | sqrt | 0.8495 |
| 8 | 10 | absolute_error | log2 | 0.8524 |
| 9 | 50 | absolute_error | sqrt | 0.8669 |
| 10 | 50 | absolute_error | log2 | 0.8607 |
| 11 | 100 | absolute_error | sqrt | 0.8675 |
| 12 | 100 | absolute_error | log2 | 0.8695 |
| 13 | 10 | poisson | sqrt | 0.8446 |
| 14 | 10 | poisson | log2 | 0.8624 |
| 15 | 50 | poisson | sqrt | 0.8663 |
| 16 | 50 | poisson | log2 | 0.8687 |
| 17 | 100 | poisson | sqrt | 0.8688 |
| 18 | 100 | poisson | log2 | 0.8685 |

**BEST MODEL WHILE HYPERTUNING IN RANDOM FOREST FOR PROBLEM**

| S.NO | N_ESTIMATORS | CRITERION | MIN_LEAF | MIN_SPLIT | R VALUE |
|---|---|---|---|---|---|
| 1 | 60 | absolute_error | 6 | 10 | 0.8912 |

The **Random Forest regression** use $R^2$ value = 0.8912